

Dual-Arm Cooperative Manipulation Through Shared Vision

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Abstract—This work presents a dual-arm manipulation system using two Kinova Gen3 Lite (6-DOF) manipulators, each with a wrist-mounted eye-in-hand camera, for coordinated target search and pick-and-place in a shared workspace. Each arm independently executes a joint-space search trajectory while running continuous fiducial (ArUco) detection on its own camera. Upon detection by either arm, the target pose is computed in the detecting arm’s own base frame from the forward kinematics of its eye-in-hand camera, and relayed into the other arm’s base frame through a fixed, measured inter-arm transform. A reachability-based rule then selects the acting arm: the detecting arm, the non-detecting arm, or, where the target lies within both reachable workspaces, whichever arm is physically closer, independent of which arm made the detection. The selected arm executes the pick-and-place sequence under damped least-squares resolved-rate control, with grasp orientation resolved under the gripper’s rotational symmetry. The complete pipeline is validated in both Gazebo/ROS2 simulation and on physical dual-arm hardware, giving a reproducible, classical-vision coordination protocol that uses no learned perception or control components. *Source code and videos for the present work are available at Git-hub and Video.*

Index Terms—Dual-Arm Manipulation, Collaborative Manipulation, Shared Vision, Resolved-Rate Control.

I. INTRODUCTION

Single-arm manipulators are the standard building block of industrial pick-and-place automation. For a fixed, well-characterized workspace, a single arm with a calibrated camera and a repeatable grasp routine is simple to deploy, easy to reason about, and reliable in operation [1]. This works well as long as every object of interest reliably falls within that one arm’s reachable envelope. As operations scale up to higher-throughput cells, larger bins and conveyors, or workspaces shared with other stations, this assumption starts to break down: a single arm’s reach is fixed, but the region objects can appear in is often larger than what one arm can physically cover. The common industrial response is to bring in a second manipulator to share the same workspace rather than scale up to a single larger (and more expensive) arm, but this immediately introduces a coordination problem [2] that a single-arm system never has to solve: when both arms can potentially act, which one actually should?

Existing work answers this in three largely distinct ways: by coupling both arms at the control level so that they jointly manipulate one object [3]; by treating assignment as a scheduling problem solved centrally from a shared, usually external, view of the workspace [4], [5]; or by settling it offline at the design stage, using reachability or capability maps to divide the workspace into static per-arm zones [6]. Sec. II takes each in turn.

Little attention has been paid to a coordination style that combines two elements, each common on its own but rarely found together: perception carried independently by each arm through its own wrist-mounted eye-in-hand camera rather than a shared external sensor, and a runtime, perception decision of which arm should act, based on live reachability computed in each arm’s own frame rather than on a precomputed zone or a centrally solved allocation. Eye-in-hand vision is well established for single-arm pick-and-place [7], and dual-arm systems have been used for perception-intensive tasks such as cooperative 3D model acquisition [8], although there the sensing comes from a stationary external depth sensor rather than each arm’s own wrist camera. In neither case is a second arm’s camera used to decide which arm should manipulate a shared target.

This paper addresses exactly that case. We present a dual-arm system in which two 6-DOF manipulators, each with its own wrist-mounted (eye-in-hand) camera, independently search a shared workspace using a fiducial marker as the target [9]. Detection by either arm’s camera is followed by a dynamic, reachability-based rule that selects which arm performs the pick, with the target pose communicated across the arms’ base frames through a fixed, measured rigid transform regardless of which arm made the detection. Manipulation itself uses classical resolved-rate control: a damped least-squares inverse-kinematics formulation for singularity-robust joint velocity control [10], [11], driven by a Cartesian pose error computed directly from live eye-in-hand visual feedback [12].

This is deliberately a small, well-scoped technique rather than a general coordination framework. We make no claim beyond it; in particular, we do not address continuous visual tracking during manipulation, multi-object scheduling, or bimanual coordinated grasping, each a separate problem in its own right. The contributions are:

- A runtime reachability-based role-assignment rule that selects the acting arm per detection, resolving workspace overlap by minimum base-to-target distance and resuming the search when neither arm can reach the target.

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- A cross-frame pose relay through a shared reference transform, which expresses a target detected by one arm's eye-in-hand camera in the other arm's base frame, enabling reachability to be evaluated for both arms without a shared external sensor.
- End-to-end validation of the full pipeline (concurrent search, cross-frame relay, role assignment, and DLS resolved-rate pick-and-place) in Gazebo/ROS2 and on physical Kinova Gen3 Lite hardware, with source code and videos released publicly.

II. RELATED WORK

Dual-arm manipulators, mirroring bi-manual human configurations, have drawn attention for sorting and object-transport applications [13], [2]. Coordination between them has been achieved by exchanging real-time kinematic data [14], sharing goal configurations over a wireless link [15], or relying on a global motion-capture system for centralized spatial tracking [16].

Early work coupled the arms at the control level, both acting on one shared object. Uchiyama and Dauchez [17] proposed a symmetric hybrid position/force scheme in which workspace force, velocity and position vectors are symmetric functions of both arms' joint-space quantities, so that neither arm dominates; the leader-follower alternative instead commands one arm's trajectory directly and derives the second arm's motion to accommodate it [18]. These and related hybrid and adaptive variants are surveyed in [3]. Throughout, coordination concerns how two arms move *together* on the same object, not which arm should act.

A second line treats coordination as discrete assignment. Behrens *et al.* [4] cast simultaneous task allocation and motion scheduling for industrial dual-arm cells as a constraint program, jointly optimizing task order and arm assignment under precedence and timing constraints. Zacharias *et al.* [6] address the same question at the design stage, building reachability and capability maps that support a static, offline division of the workspace into per-arm zones. In both, assignment is solved centrally by an optimizer or fixed in advance by geometry, rather than decided at runtime from live sensing.

A third line learns the assignment. Jermann *et al.* [5] apply reinforcement learning to coordinate multiple arms for pick-and-place sorting; Alles and Aljalbout [19] couple two single-arm controllers through a centrally learned policy for peg-in-hole assembly; and Chen *et al.* [20] predict object-centric motion flows with a vision-language-model-assisted diffusion framework, dynamically assigning them to whichever arm should execute them. Across this line, the low-level motion and the question of which arm acts are increasingly folded into a single learned model.

The present protocol differs from all three: it keeps the kinematics classical and explicitly modeled (Sec. III-B), and evaluates reachability at runtime for each detection (Sec. III-F) from pose information relayed between the arms' own cameras, rather than from a shared external sensor, an offline zone division [6], or a learned policy.

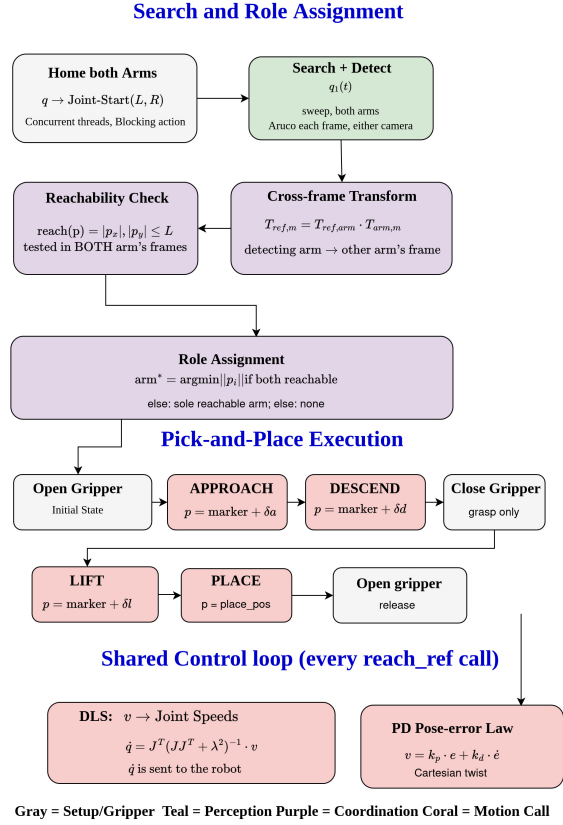


Fig. 1. Overall Architecture: concurrent homing and search, cross-frame pose relay, reachability-based role assignment, and pick-and-place execution under the shared PD-into-DLS control loop.

III. METHODOLOGY

A. System Architecture

The platform consists of two Kinova Gen3 Lite manipulators (6 DOF each), denoted $\mathcal{A}_L$ (LEFT) and $\mathcal{A}_R$ (RIGHT), each with a wrist-mounted (eye-in-hand) RGB camera and sharing a common tabletop workspace. Fig. 1 summarizes the pipeline. Its five components, the search trajectory, the cross-frame pose relay, the reachability-based role-assignment rule, the manipulation sequence, and the underlying resolved-rate control law, are treated in turn in the subsections below; every Cartesian motion stage is driven by the same PD-law-into-damped-least-squares loop.

B. Kinematic Model

Each arm's forward kinematics is a product of single-DOF joint-dependent homogeneous transforms with fixed link geometry, extended by two further fixed transforms to the tool frame and the eye-in-hand camera frame, as in Eq. 1:

$$\begin{aligned} T_0^6(\mathbf{q}) &= T_0^1(q_1) T_1^2(q_2) T_2^3(q_3) T_3^4(q_4) T_4^5(q_5) T_5^6(q_6), \\ T_0^e(\mathbf{q}) &= T_0^6(\mathbf{q}) T_6^e, \quad T_0^c(\mathbf{q}) = T_0^e(\mathbf{q}) T_e^c, \end{aligned} \quad (1)$$

with $\mathbf{q} = [q_1, \dots, q_6] \in \mathbb{R}^6$, T_6^e the fixed tool-frame offset, and T_e^c the fixed eye-in-hand camera offset (identical, by construction, on both arms). The manipulator Jacobian

$J(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$ is obtained by central-difference numerical differentiation of T_0^e as in Eq. 2:

$$\mathbf{J}_{:,i} = \begin{bmatrix} \frac{\mathbf{p}(\mathbf{q} + \epsilon \mathbf{e}_i) - \mathbf{p}(\mathbf{q} - \epsilon \mathbf{e}_i)}{2\epsilon} \\ \frac{\log(\mathbf{R}(\mathbf{q} + \epsilon \mathbf{e}_i)\mathbf{R}(\mathbf{q} - \epsilon \mathbf{e}_i)^T)}{2\epsilon} \end{bmatrix}, \quad \epsilon = 10^{-6}. \quad (2)$$

where $\mathbf{p}(\cdot)$, $\mathbf{R}(\cdot)$ are the translational and rotational parts of T_0^e and $\mathbf{e}_i$ is the i -th standard basis vector. Rotations do not subtract meaningfully, so the angular rows use the matrix logarithm $\log(\cdot)$, which maps the relative rotation $\mathbf{R}(\mathbf{q} + \epsilon \mathbf{e}_i)\mathbf{R}(\mathbf{q} - \epsilon \mathbf{e}_i)^T \in SO(3)$ (orthogonality giving $\mathbf{R}^{-1} = \mathbf{R}^T$) to its axis-angle vector in $\mathbb{R}^3$. That vector is accumulated over $[-\epsilon, \epsilon]$, so dividing by 2ϵ recovers the angular-velocity column with the same $\mathcal{O}(\epsilon^2)$ central-difference accuracy as the linear rows.

C. Search Trajectory

Both arms first execute a blocking angular Action move (in independent threads, so neither arm idles waiting for the other) to a per-arm predefined start configuration $\mathbf{q}_0^L, \mathbf{q}_0^R$.

Each arm then independently sweeps its own joint 1 at constant angular speed, reversing direction at each bound rather than following a smooth periodic profile. The sweep is therefore a constant-speed bang-bang (triangle-wave) trajectory in q_1 , governed by a direction state $s(t) \in \{-1, +1\}$ that flips only when a bound is crossed, as in Eq. 3:

$$\begin{aligned} \dot{q}_1(t) &= s(t) \dot{q}_{\text{sweep}}, \\ s(t^+) &= \begin{cases} -1, & q_1(t) \geq q_1^{\max}, \quad s(t) = +1, \\ +1, & q_1(t) \leq q_1^{\min}, \quad s(t) = -1, \\ s(t), & \text{otherwise.} \end{cases} \end{aligned} \quad (3)$$

with $s(0) = +1$. The hysteresis on s is what makes Eq. 3 well posed: inside the band $[q_1^{\min}, q_1^{\max}]$ the direction is held, and it is reassigned only at a bound. The bounds themselves are computed from that arm's own start angle $q_{1,0}$, a sweep half-range Δ_{search} , and the joint's hard limits $q_1^{\text{lo}}, q_1^{\text{hi}}$ with a safety margin δ_{lim} , as in Eq. 4:

$$\begin{aligned} q_1^{\min} &= \max(q_{1,0} - \Delta_{\text{search}}, q_1^{\text{lo}} + \delta_{\text{lim}}), \\ q_1^{\max} &= \min(q_{1,0} + \Delta_{\text{search}}, q_1^{\text{hi}} - \delta_{\text{lim}}). \end{aligned} \quad (4)$$

The margin δ_{lim} is distinct from the central-difference step ϵ of Eq. 2; the two are unrelated quantities. While sweeping, each arm continuously runs ArUco marker detection on its own camera stream.

D. Visual Target Localization

Given a detected marker's corner correspondences, a PnP-based pose estimator yields $(\mathbf{t}_{cm}, R_{cm})$ relative to the camera. Using the live camera pose $T_0^c(\mathbf{q})$ of Eq. 1, evaluated at the joint configuration held at the instant of detection, the marker pose is expressed in the detecting arm's own base frame as in Eq. 5:

$$T_{\text{arm}}^m = T_0^c(\mathbf{q}) T_c^m, \quad T_c^m = \begin{bmatrix} R_{cm} & \mathbf{t}_{cm} \\ \mathbf{0}^T & 1 \end{bmatrix}. \quad (5)$$

If more than one marker is visible in a frame, the one with the smallest $\|\mathbf{t}_{cm}\|$ is taken as the target.

E. Cross-Frame Pose Communication

The two arms' base frames are related by a fixed rigid transform $T_{\text{ref}}^R \in SE(3)$, obtained from the measured mounting geometry of the rig and held constant thereafter (LEFT is the shared reference frame, so $T_{\text{ref}}^L = I$). The detected marker pose T_{arm}^m of Eq. 5 is relayed into the shared frame and then into the other arm's own frame by Eq. 6:

$$\begin{aligned} T_{\text{ref}}^m &= T_{\text{ref}}^{\text{arm}} T_{\text{arm}}^m, \\ T_{\text{arm}'}^m &= \left(T_{\text{ref}}^{\text{arm}'}\right)^{-1} T_{\text{ref}}^m. \end{aligned} \quad (6)$$

This round-trip through the shared frame is what allows a target detected by one arm's camera to be evaluated in the other arm's own frame, regardless of which arm made the detection.

F. Reachability-Based Role Assignment

Reachability of a position $\mathbf{p}$ (expressed in a given arm's own base frame) is tested against that arm's predefined workspace half-extent L_w , as in Eq. 7. The test bounds the target's x and y coordinates only and places no constraint on z :

$$\text{reach}(\mathbf{p}) = \begin{cases} 1, & |p_x| \leq L_w \text{ and } |p_y| \leq L_w, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

Upon detection, $\text{reach}(\cdot)$ is evaluated for both arms using their respective local poses obtained from Eq. 6 (Sec. III-E), and the acting arm is selected as in Eq. 8:

$$\text{arm}^* = \begin{cases} a_d, & \text{reach}(\mathbf{p}_{a_d}) \wedge \neg \text{reach}(\mathbf{p}_{a_o}), \\ a_o, & \text{reach}(\mathbf{p}_{a_o}) \wedge \neg \text{reach}(\mathbf{p}_{a_d}), \\ \arg \min_{i \in \{a_d, a_o\}} \|\mathbf{p}_i\|, & \text{reach}(\mathbf{p}_{a_d}) \wedge \text{reach}(\mathbf{p}_{a_o}), \\ \emptyset, & \text{otherwise.} \end{cases} \quad (8)$$

where a_d and a_o denote the detecting and the other arm respectively, $\mathbf{p}_i$ is the marker position expressed in arm i 's own base frame (so the two vectors compared in the third branch live in different frames, each measuring that arm's own base-to-target distance), and $\emptyset$ denotes that no arm is assigned, in which case the sweep of Eq. 3 resumes and the next detection is evaluated afresh.

G. Manipulation Sequence

Once arm^* is selected by Eq. 8, it executes a seven-step sequence: open gripper; APPROACH; DESCEND; close gripper (grasp); LIFT; PLACE; open gripper (release). Only APPROACH, DESCEND, and LIFT are derived from the marker position $\mathbf{p}_m$ (in arm^* 's own frame); each is *chained* from the previous target rather than independently offset from the marker, and grasp itself involves no motion target,

TABLE I
SYSTEM AND CONTROLLER PARAMETERS.

Symbol	Description	Value
<i>Kinematics and rig geometry</i>		
T_{ref}^R	Inter-arm transform	$[1.08, 0, 0]$ m, $R_z(180^\circ)$
L_w	Workspace half-extent (Eq. 7)	0.5 m
<i>Search trajectory</i>		
$\dot{q}_{\text{sweep}}$	Joint-1 sweep speed	$8^\circ/\text{s}$
Δ_{search}	Sweep half-range	120°
δ_{lim}	Joint-limit margin (Eq. 4)	0.08 rad
<i>Perception</i>		
–	ArUco dictionary	5X5.50
–	Marker side length	0.05 m
<i>Manipulation waypoints (Eq. 9)</i>		
$\mathbf{a}$	Approach offset, $\mathcal{A}_L$	$[0.05, 0.02, 0.07]$ m
$\mathbf{a}$	Approach offset, $\mathcal{A}_R$	$[0.03, 0.07, 0.07]$ m
d	Descend distance	0.08 m
l	Lift height	0.10 m
(ϕ^*, θ^*)	Grasp roll/pitch	$0^\circ, -180^\circ$
<i>Resolved-rate control</i>		
$K_p^{\text{pos}}, K_d^{\text{pos}}$	Position PD gains	2.0, 0.05
$K_p^{\text{ori}}, K_d^{\text{ori}}$	Orientation PD gains	0.9, 0.04
τ	Deriv. filter const. (Eq. 11)	0.06 s
$\lambda_{\text{nom}}, \lambda_{\text{sing}}$	DLS damping (Eq. 13)	0.12, 0.45
w_{min}	Manip. threshold (Eq. 13)	0.01
$\dot{q}_{\text{max}}$	Joint-speed saturation	0.2 rad/s
Δt	Control loop period (10 Hz)	0.1 s
<i>Convergence tolerances (pos. / ori.)</i>		
–	APPROACH, LIFT, PLACE	8 mm / 5°
–	DESCEND (grasp)	5 mm / 3°
–	Consecutive cycles in tolerance	3

as in Eq. 9:

$$\begin{aligned}
 \mathbf{p}_{\text{approach}} &= \mathbf{p}_m + \mathbf{a}, \\
 \mathbf{p}_{\text{descend}} &= \mathbf{p}_{\text{approach}} - \begin{bmatrix} 0 \\ 0 \\ d \end{bmatrix}, \\
 \mathbf{p}_{\text{lift}} &= \mathbf{p}_{\text{descend}} + \begin{bmatrix} 0 \\ 0 \\ l \end{bmatrix}.
 \end{aligned} \tag{9}$$

where $\mathbf{a}$ is a fixed per-arm approach offset and d, l are fixed scalar descend/lift distances (grasp is a gripper actuation at $\mathbf{p}_{\text{descend}}$, not a fifth target). The PLACE target $\mathbf{p}_{\text{place}}$ is, by contrast, a fixed pre-configured drop-off pose for arm * , independent of $\mathbf{p}_m$.

Every target from Eq. 9 is additionally clamped to a hard floor $z \geq z_{\text{min}} = 0$ in the acting arm's own base frame, so that no commanded tool pose can drive the fingers into the tabletop. The three marker-derived targets share a common grasp orientation: a fixed top-down roll and pitch (ϕ^*, θ^*) with the yaw taken from the marker. That yaw is resolved under the gripper's 180° rotational symmetry by choosing, of the two equivalent orientations, whichever has the greater quaternion alignment with the arm's current tool orientation, minimizing unnecessary wrist rotation.

H. Resolved-Rate Motion Control

Every one of the four motion targets above (APPROACH, DESCEND, LIFT, PLACE) is reached by the same control loop. Given the position/orientation error $(\mathbf{e}_p, \mathbf{e}_o, \theta_e)$ between the current and target tool pose, a PD law with a first-order low-pass filtered derivative produces a Cartesian

twist $\mathbf{v}$ (an intermediate quantity, not a robot command), as in Eq. 10:

$$\mathbf{v} = \begin{bmatrix} K_p^{\text{pos}} \mathbf{e}_p + K_d^{\text{pos}} \dot{\mathbf{e}}_{p,\text{filt}} \\ K_p^{\text{ori}} \mathbf{e}_o + K_d^{\text{ori}} \dot{\mathbf{e}}_{o,\text{filt}} \end{bmatrix}. \tag{10}$$

The filtered derivative in Eq. 10 is a discrete first-order low-pass of the raw error difference, with time constant τ and control period Δt , as in Eq. 11:

$$\dot{\mathbf{e}}_{\text{filt}} \leftarrow \alpha \frac{\mathbf{e}_k - \mathbf{e}_{k-1}}{\Delta t} + (1 - \alpha) \dot{\mathbf{e}}_{\text{filt}}, \quad \alpha = \frac{\Delta t}{\tau + \Delta t}. \tag{11}$$

applied independently to $\mathbf{e}_p$ and $\mathbf{e}_o$; without it, the K_d terms would amplify the quantization noise of the 10 Hz joint feedback straight into the commanded twist. The twist is mapped to the joint speeds actually sent to the arm via damped least-squares (DLS) inverse differential kinematics, Eq. 12:

$$\dot{\mathbf{q}} = J^T (J J^T + \lambda^2 I)^{-1} \mathbf{v}. \tag{12}$$

The damping λ in Eq. 12 is not constant: it is switched between a nominal value λ_{nom} and a larger value λ_{sing} whenever the arm approaches a singularity, detected through the manipulability measure $w(\mathbf{q})$ falling below a threshold w_{min} , as in Eq. 13:

$$\lambda(\mathbf{q}) = \begin{cases} \lambda_{\text{sing}}, & w(\mathbf{q}) < w_{\text{min}}, \\ \lambda_{\text{nom}}, & \text{otherwise}, \end{cases} \quad w(\mathbf{q}) = \sqrt{\det(J J^T)}. \tag{13}$$

Raising λ near a singularity trades tracking accuracy for a bounded joint-speed solution, which is the behaviour wanted there. The resulting $\dot{\mathbf{q}}$ is scaled down near joint limits and saturated at $\dot{q}_{\text{max}}$ before being sent.

Position error is a plain vector difference between the target and current tool position, $\mathbf{e}_p = \mathbf{p}^* - \mathbf{p}(\mathbf{q})$. Orientation error requires more care, since rotations cannot simply be subtracted. The relative error quaternion is formed and sign-corrected so the controller always takes the shorter rotational path:

$$\begin{aligned}
 \mathbf{q}_e &= \mathbf{q}_{\text{tgt}} \otimes \mathbf{q}_{\text{curr}}^{-1}, \\
 \mathbf{q}_e &\leftarrow -\mathbf{q}_e, \quad \text{if } q_e^{(4)} < 0.
 \end{aligned} \tag{14}$$

where $\mathbf{q}_{\text{tgt}}$ and $\mathbf{q}_{\text{curr}}$ are the target and current tool orientations as unit quaternions, $\otimes$ is quaternion multiplication, and $q_e^{(4)}$ is the scalar (w) component. The sign flip in Eq. 14 exploits the fact that $\mathbf{q}_e$ and $-\mathbf{q}_e$ denote the same rotation, removing the ambiguity that would otherwise let the controller rotate the long way around. Writing $\mathbf{q}_e^{(1:3)}$ for its vector part, the error used by the PD law of Eq. 10 is:

$$\mathbf{e}_o = 2 \mathbf{q}_e^{(1:3)}. \tag{15}$$

and the residual rotation angle, used only to test convergence and never fed into the control law, is recovered exactly via Eq. 16:

$$\theta_e = 2 \operatorname{atan2} \left(\left\| \mathbf{q}_e^{(1:3)} \right\|, q_e^{(4)} \right). \tag{16}$$

Eq. 15 follows from $\mathbf{q}_e = (\sin(\theta_e/2) \hat{\mathbf{n}}, \cos(\theta_e/2))$, $\hat{\mathbf{n}}$ being the rotation axis: doubling the vector part gives

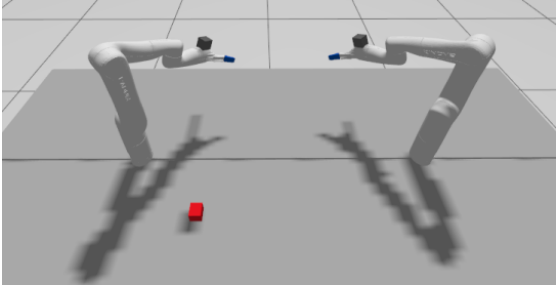


Fig. 2. Gazebo/ROS2 simulation setup: two Kinova Gen3 Lite arms sharing a tabletop workspace with a fiducial target.

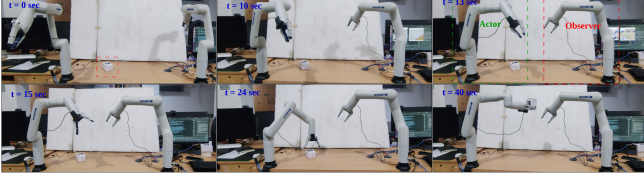


Fig. 3. Experiment 1 (hardware): LEFT arm selected as actor, RIGHT arm idle. Frames show search, detection, approach, grasp, and place.

$2 \sin(\theta_e/2) \hat{n} \approx \theta_e \hat{n}$ for small errors, the familiar axis-scaled-by-angle vector. atan2 is used in Eq. 16 for numerical stability across the full range of θ_e , including near 0° and 180° . Convergence is declared once $\|e_p\|$ and θ_e (Eq. 16) both remain within the per-waypoint tolerances listed in Table I for three consecutive control cycles, after which the next waypoint's target is computed from Eq. 9 and the loop repeats. The DESCEND waypoint, being the one at which the grasp is executed, is held to a tighter tolerance than the others.

IV. RESULTS AND DISCUSSION

The pipeline was implemented both in a Gazebo/ROS2 simulation of the rig (Fig. 2) and on the physical Kinova Gen3 Lite hardware (Fig. 3).

Two representative hardware trials illustrate the complete pipeline end-to-end. In Experiment 1 the LEFT arm was selected as the acting arm (Fig. 3, Fig. 4, Fig. 5); Experiment 2 is the mirrored case, with the marker placed so that RIGHT was selected instead. In both, the non-selected arm remained

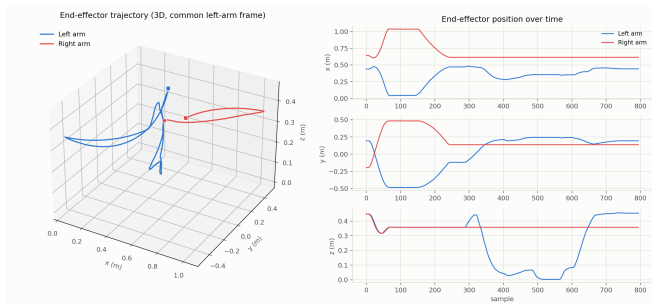


Fig. 4. Experiment 1: end-effector trajectories in the common LEFT-arm frame (left) and per-axis position over time (right). LEFT traces approach–descend–lift–place; RIGHT remains stationary.

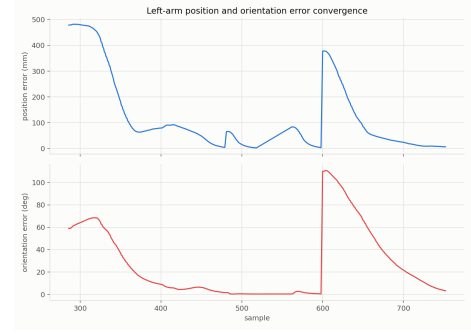


Fig. 5. Experiment 1: LEFT-arm position and orientation error, showing a convergence transient at each waypoint transition.

TABLE II

TERMINAL POSITION/ORIENTATION ERROR AT DECLARED CONVERGENCE (THE CONTROLLER'S OWN FORWARD-KINEMATIC ERROR AT THE FINAL WAYPOINT), FIVE TRIALS PER ACTING ARM.

Trial	LEFT acting		RIGHT acting	
	Pos (mm)	Orn ($^\circ$)	Pos (mm)	Orn ($^\circ$)
1	2.55	0.31	2.18	0.84
2	2.55	0.78	2.64	0.77
3	4.59	0.05	3.14	0.64
4	3.26	1.22	2.45	1.02
5	2.82	0.72	3.40	0.16
Mean	3.15	0.62	2.76	0.69
S.D.	0.85	0.45	0.50	0.33

idle, and the marker fell inside exactly one arm's workspace box, so it is the sole-reachable-arm branch of Eq. 8 that was exercised. The overlap tie-break and the resume-search branches are discussed as limitations in Sec. V.

Fig. 4 shows the LEFT arm's end-effector tracing the full waypoint sequence approach, descend, lift, and place, while the RIGHT arm's position remains essentially flat across the same sample range, consistent with it never receiving a motion command once the reachability-based rule assigned the task to LEFT. The corresponding error convergence in Fig. 5 shows the position and orientation error being driven from its initial value toward zero and then stepping back up at each waypoint transition before converging again. This sawtooth pattern is expected: each of the four motion targets of Eq. 9 presents the resolved-rate controller with a fresh, non-zero error that must be driven back down under the same PD-into-DLS loop, so the plot shows a sequence of independent convergence transients rather than the tracking of a single continuous trajectory.

Experiment 2 reproduces the same qualitative pattern with the roles reversed: a stationary idle arm, and successive convergence transients on the acting arm. The pipeline therefore behaves consistently regardless of which arm the reachability rule selects, rather than one arm's kinematics or camera calibration producing systematically different convergence behavior. Per-trial errors for both cases are given in Table II.

Table II reports the terminal position and orientation error at the instant convergence was declared on the final waypoint, across five trials with **Left** selected as the acting arm and five

trials with **Right** selected. Mean terminal position error was 3.153 mm (**Left**) and 2.762 mm (**Right**), with orientation error of 0.616° (**Left**) and 0.686° (**Right**); sample standard deviations were under 1 mm and under 0.5° in both cases.

Two caveats govern how these numbers should be read. First, they are the *controller's own* terminal pose error, computed from forward kinematics at the moment the convergence test of Sec. III-H was satisfied, and are therefore bounded by construction by the tolerances in Table I. They establish that the loop terminates reliably, and comparably tightly for either arm, but not the absolute placement accuracy, which would require independent metrology. Second, with five trials per configuration and no hypothesis test, the near-equal means support only the claim that *no systematic asymmetry was observed* between the arms, despite their different mounting positions and approach offsets. These trials establish repeatability at a proof-of-concept level, not a fully powered statistical characterization.

V. CONCLUSION

We presented a dual-arm system in which two Kinova Gen3 Lite manipulators, each with its own eye-in-hand camera, search a shared workspace independently and cooperate through a reachability-based role-assignment rule, with the target pose relayed between their base frames regardless of which arm detected it. Validation on hardware gave terminal forward-kinematic errors below 5 mm and 1.3° over five trials per acting arm, with no systematic asymmetry observed between the arms.

Four limitations bound this result. The trials are few, and the reported errors are the controller's own convergence errors, not independently measured placement accuracy. The reachability test (Eq. 7) is an axis-aligned bound on x and y alone, with no manipulability or collision-awareness, and the overlap tie-break uses plain Euclidean distance rather than motion cost. Both hardware trials fell into the sole-reachable-arm branch of Eq. 8, so the overlap and resume-search branches, though implemented, remain to be demonstrated. Finally, role assignment is decided once, at detection: the system does not monitor whether the acting arm's view of the target stays usable during manipulation.

That last point is the most direct avenue for future work. Letting the idle arm keep observing the target and relay a corrected pose if the acting arm's view degrades mid-task would turn a one-time handoff into a continuously monitored decision. Beyond this, natural extensions are a larger trial set, a manipulability-weighted reachability measure, and multi-object scheduling across the two arms.

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